

PINNs for optimal control

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1 Problem formulation

Let $\Omega \subset \mathbb{R}^d$ ($d = 1, 2$) be a bounded domain with sufficiently smooth boundary $\partial\Omega$, let $T > 0$, and set $Q := \Omega \times (0, T)$ and $\Sigma := \partial\Omega \times (0, T)$. We seek a state $y : Q \rightarrow \mathbb{R}$ and a Dirichlet boundary control $u : \Sigma \rightarrow \mathbb{R}$ minimizing

$$J(y, u) = \frac{1}{2} \int_Q |y - y_d|^2 dx dt + \frac{\alpha}{2} \int_\Sigma |u|^2 ds dt, \quad (1)$$

subject to

$$\begin{cases} \partial_t y - \nu \Delta y + f(y) = g, & \text{in } Q, \\ y = u, & \text{on } \Sigma, \\ y(x, 0) = y_0(x), & \text{in } \Omega. \end{cases} \quad (2)$$

Here $\nu > 0$ is the diffusion coefficient, g is a prescribed source term, $y_d \in L^2(Q)$ is the desired trajectory, and $\alpha > 0$ is the Tikhonov regularization parameter. The source g is fixed problem data and is not an optimization variable. The case $g = 0$ is included. The boundary condition is of Dirichlet type, so the control acts directly on the boundary trace of the state.

2 Methodology

We compare a direct PINN, which minimizes a penalized form of the constrained objective, with an indirect PINN, which solves the first-order optimality system. Both methods use a single control network $U_\psi(x, t)$. To keep the residual notation common to both formulations, define the state-equation residual operator

$$\text{Res}_y[v] := \partial_t v - \nu \Delta v + f(v) - g. \quad (3)$$

2.1 Direct PINN Formulation

The state and control are approximated by $Y_\theta(x, t)$ and $U_\psi(x, t)$. Their state-equation residual is $\mathcal{R}_y := \text{Res}_y[Y_\theta]$. The direct loss is

$$\mathcal{L}_{\text{dir}} = w_{\text{res}} \mathcal{L}_{\text{res}}^y + w_{\text{bc}} \mathcal{L}_{\text{bc}}^y + w_{\text{ic}} \mathcal{L}_{\text{ic}}^y + \frac{1}{2} \mathcal{L}_{\text{track}} + \frac{\alpha}{2} \mathcal{L}_{\text{reg}}, \quad (4)$$

where

$$\mathcal{L}_{\text{res}}^y := \frac{1}{N_{\text{int}}} \sum_{j=1}^{N_{\text{int}}} |\mathcal{R}_y(x_j, t_j)|^2, \quad (5)$$

$$\mathcal{L}_{\text{bc}}^y := \frac{1}{N_{\text{bc}}} \sum_{j=1}^{N_{\text{bc}}} |Y_\theta(x_j^b, t_j^b) - U_\psi(x_j^b, t_j^b)|^2, \quad (6)$$

$$\mathcal{L}_{\text{ic}}^y := \frac{1}{N_{\text{ic}}} \sum_{i=1}^{N_{\text{ic}}} |Y_\theta(x_i, 0) - y_0(x_i)|^2, \quad (7)$$

$$\mathcal{L}_{\text{track}} := \frac{1}{N_{\text{int}}} \sum_{j=1}^{N_{\text{int}}} |Y_\theta(x_j, t_j) - y_d(x_j, t_j)|^2. \quad (8)$$

In one space dimension the boundary consists of two endpoints, and the control regularization uses counting measure on $\partial\Omega$:

$$\mathcal{L}_{\text{reg}} := \frac{1}{N_{\text{bc}}} \sum_{j=1}^{N_{\text{bc}}} (|U_\psi(0, t_j^b)|^2 + |U_\psi(1, t_j^b)|^2). \quad (9)$$

This formulation is simple and avoids an adjoint network, but boundary satisfaction depends on the penalty weight and can be delicate [1].

2.2 Indirect PINN Formulation with Hard Constraints

2.2.1 First-order optimality system

With

$$\mathcal{L}(y, u, \lambda) = J(y, u) + \int_Q \lambda (\partial_t y - \nu \Delta y + f(y) - g) \, dx \, dt, \quad (10)$$

integration by parts gives

$$\partial_t y - \nu \Delta y + f(y) - g = 0 \quad \text{in } Q, \quad (11)$$

$$-\partial_t \lambda - \nu \Delta \lambda + f'(y) \lambda + (y - y_d) = 0 \quad \text{in } Q, \quad (12)$$

$$\alpha u + \nu \partial_n \lambda = 0 \quad \text{on } \Sigma, \quad (13)$$

$$y = u, \quad \lambda = 0 \quad \text{on } \Sigma, \quad (14)$$

$$y(x, 0) = y_0(x), \quad \lambda(x, T) = 0 \quad \text{in } \Omega. \quad (15)$$

For $\Omega = (0, 1)$, $\partial_n \lambda(0, t) = -\lambda_x(0, t)$ and $\partial_n \lambda(1, t) = \lambda_x(1, t)$.

2.2.2 One control network and hard constraints

We use one network $U_\psi(x, t)$ and define its two boundary traces by

$$u_0(t) := U_\psi(0, t), \quad u_1(t) := U_\psi(1, t). \quad (16)$$

For $\Omega = (0, 1)$, the boundary lifting and state trial function are

$$B_\psi(x, t) := (1 - x)U_\psi(0, t) + xU_\psi(1, t), \quad (17)$$

$$Y_\theta(x, t) := B_\psi(x, t) + x(1 - x)N_\theta(x, t). \quad (18)$$

Thus $Y_\theta(0, t) = U_\psi(0, t)$ and $Y_\theta(1, t) = U_\psi(1, t)$ for every parameter value. The adjoint is parameterized as

$$\Lambda_\phi(x, t) := x(1 - x)(T - t)M_\phi(x, t), \quad (19)$$

which enforces both $\Lambda_\phi = 0$ on the spatial boundary and $\Lambda_\phi(x, T) = 0$. The state initial condition remains a soft constraint.

2.2.3 Indirect PINN loss

Define

$$\mathcal{R}_y := \text{Res}_y[Y_\theta], \quad (20)$$

$$\mathcal{R}_\lambda := -\partial_t \Lambda_\phi - \nu \Delta \Lambda_\phi + f'(Y_\theta) \Lambda_\phi + Y_\theta - y_d, \quad (21)$$

$$\mathcal{R}_u := \alpha U_\psi + \nu \partial_n \Lambda_\phi \quad \text{on } \Sigma. \quad (22)$$

In one space dimension, its two traces are

$$\mathcal{R}_{u,0}(t) := \mathcal{R}_u(0, t) = \alpha U_\psi(0, t) - \nu \partial_x \Lambda_\phi(0, t), \quad \mathcal{R}_{u,1}(t) := \mathcal{R}_u(1, t) = \alpha U_\psi(1, t) + \nu \partial_x \Lambda_\phi(1, t). \quad (23)$$

The indirect loss is

$$\mathcal{L}_{\text{ind}} = w_y \mathcal{L}_y + w_\lambda \mathcal{L}_\lambda + w_u \mathcal{L}_u + w_{\text{ic}} \mathcal{L}_{\text{ic}}, \quad (24)$$

with

$$\mathcal{L}_y := \frac{1}{N_{\text{int}}} \sum_j |\mathcal{R}_y(x_j, t_j)|^2, \quad (25)$$

$$\mathcal{L}_\lambda := \frac{1}{N_{\text{int}}} \sum_j |\mathcal{R}_\lambda(x_j, t_j)|^2, \quad (26)$$

$$\mathcal{L}_u := \frac{1}{2N_{\text{bc}}} \sum_j (|\mathcal{R}_{u,0}(t_j^b)|^2 + |\mathcal{R}_{u,1}(t_j^b)|^2), \quad (27)$$

$$\mathcal{L}_{\text{ic}} := \frac{1}{N_{\text{ic}}} \sum_i |Y_\theta(x_i, 0) - y_0(x_i)|^2. \quad (28)$$

Thus $\mathcal{L}_u$ is the sample mean with respect to time and the normalized counting measure on the two endpoints; the factor $1/2$ is part of its definition throughout the paper. There are no state-boundary, adjoint-boundary, or adjoint-terminal penalties because these constraints are satisfied by construction.

3 Error Analysis

3.1 Standing Assumptions

The estimates in this section are conditional on the following standing assumptions, which we state explicitly rather than leaving implicit.

- (A1) **Well-posedness.** The optimality system (11)–(15) admits a unique solution (y^*, u^*, λ^*) in an appropriate energy class, and the linearization of this system about (y^*, λ^*) satisfies an inf-sup (stability) condition with constant C_{stab} .
- (A2) **Regularity of the Dirichlet control problem.** Dirichlet boundary control of a parabolic equation is, in general, a low-regularity problem: for boundary data $u \in L^2(\Sigma)$ the state y is typically only a *very weak solution* (transposition solution), and the trace $\partial_n \lambda|_\Sigma$ appearing in (13) need not exist in the classical sense [? ?]. We assume throughout that the data (y_0, y_d, u^*) are smooth enough, and Ω is regular enough, that (y^*, λ^*) lie in $H^{2,1}(Q)$ -type spaces in which all traces below are classically well defined. This is a genuine restriction, not a technicality; results that require only $u \in L^2(\Sigma)$ are outside the scope of the estimates below.
- (A3) **Nonlinearity.** $f \in C^1(\mathbb{R})$ and f' is bounded below, $f'(s) \geq -c_f$ for some $c_f \geq 0$ (satisfied by the Allen–Cahn nonlinearity $f(y) = y^3 - y$, for which $f'(s) = 3s^2 - 1 \geq -1$).
- (A4) **Approximation capacity.** The network architectures for $Y_\theta, U_\psi, \Lambda_\phi$ are rich enough that the best-approximation error ϵ_{net} defined in (31) can be made arbitrarily small by increasing width/depth. No explicit rate in network size is claimed here; this is treated as an assumption, not a proven property of the architecture.

3.2 Decomposition of the Total Error

Let $e_y := y^* - Y_\theta$, $e_\lambda := \lambda^* - \Lambda_\phi$, $e_u := u^* - U_\psi$ denote the errors in the state, adjoint, and control, relative to the exact solution of the optimality system (not to a discretized/mesh solution). The total error is measured in the norm

$$\|(e_y, e_\lambda, e_u)\|_{\mathcal{E}}^2 := \|e_y\|_{L^2(0,T;H^1(\Omega))}^2 + \|e_\lambda\|_{L^2(0,T;H^1(\Omega))}^2 + \|e_u\|_{L^2(\Sigma)}^2. \quad (29)$$

Using the triangle inequality and the stability assumption (A1), the total error can be decomposed into three main contributions:

$$\|(e_y, e_\lambda, e_u)\|_{\mathcal{E}} \lesssim \mathcal{E}_{\text{approx}} + \mathcal{E}_{\text{opt}} + \mathcal{E}_{\text{quad}}, \quad (30)$$

where $\mathcal{E}_{\text{approx}}$ is the best-approximation error of the neural network architecture (item (A4)), $\mathcal{E}_{\text{opt}}$ is the optimization error, i.e. how far the trained loss (24) is from its infimum over the chosen architecture, and $\mathcal{E}_{\text{quad}}$ is the quadrature (sampling) error between the Monte-Carlo loss $\mathcal{L}_{\text{ind}}$ and the continuous residual norms it estimates. We do not list a separate boundary or terminal contribution because these conditions are hard-constrained; the state initial mismatch is retained explicitly in the trainable loss and in the estimator of Section 3.4.

3.3 A Priori Error Estimate

Theorem 1 (Conditional a priori estimate). *Under Assumptions (A1)–(A3), let (y^*, u^*, λ^*) be the exact solution of (11)–(15). Suppose the networks Y_θ , U_ψ , Λ_ϕ can approximate (y^*, λ^*, u^*) to accuracy ϵ_{net} in the $\mathcal{E}$ -norm,*

$$\inf_{\theta, \psi, \phi} \|(y^* - Y_\theta, \lambda^* - \Lambda_\phi, u^* - U_\psi)\|_{\mathcal{E}} \leq \epsilon_{\text{net}}, \quad (31)$$

and suppose training reaches parameters $(\theta^*, \psi^*, \phi^*)$ with continuous residual norm

$$\|\mathcal{R}_y\|_{L^2_{\text{av}}(Q)}^2 + \|\mathcal{R}_\lambda\|_{L^2_{\text{av}}(Q)}^2 + \|\mathcal{R}_u\|_{L^2_{\text{av}}(\Sigma)}^2 \leq \epsilon_{\text{loss}}^2. \quad (32)$$

Then

$$\|(y^* - Y_{\theta^*}, \lambda^* - \Lambda_{\phi^*}, u^* - U_{\psi^*})\|_{\mathcal{E}} \lesssim \epsilon_{\text{net}} + C_{\text{stab}} \epsilon_{\text{loss}}, \quad (33)$$

with C_{stab} the stability constant of Assumption (A1).

Sketch of proof. Write $e = (e_y, e_\lambda, e_u)$ for the error at the trained parameters, and split $e = e^{\text{approx}} + e^{\text{res}}$, where e^{approx} is the error of the *best* network approximation (controlled by ϵ_{net} via (A4)) and $e^{\text{res}} := (y^* - Y_{\theta^*}, \dots) - e^{\text{approx}}$ is what remains. By linearity, e^{res} solves the linearization of (11)–(13) about (y^*, λ^*) with right-hand side equal to $(\mathcal{R}_y, \mathcal{R}_\lambda, \mathcal{R}_u)$ evaluated at the trained networks (this uses (A2) to make the linearization and its traces meaningful, and (A3) to control the sign of the reaction term $f'(y^*)$ so that the linearized adjoint operator is coercive up to a lower-order term, via a standard exponential-weight / Gårding-inequality argument). The stability assumption (A1) and the equivalence in (??) then give $\|e^{\text{res}}\|_{\mathcal{E}} \leq C_{\text{stab}} (\|\mathcal{R}_y\|_{L^2_{\text{av}}(Q)} + \|\mathcal{R}_\lambda\|_{L^2_{\text{av}}(Q)} + \|\mathcal{R}_u\|_{L^2_{\text{av}}(\Sigma)}) \leq C_{\text{stab}} \epsilon_{\text{loss}}$, after absorbing the fixed domain factors into C_{stab} . Combining with $\|e^{\text{approx}}\|_{\mathcal{E}} \leq \epsilon_{\text{net}}$ and the triangle inequality gives (33). We emphasize that this is a *sketch*: a full proof requires making (A1)–(A2) precise (function spaces, trace theorems) for the specific f and domain considered, which is not carried out here. $\square$

Remark 1. Bound (33) uses the continuous residual norm ϵ_{loss} , not the Monte-Carlo loss $\mathcal{L}_{\text{ind}}$ that is actually minimized during training. The gap between the two is exactly $\mathcal{E}_{\text{quad}}$ in (30); see Section 3.5 for how it is controlled in practice.

Remark 2. For the Allen–Cahn nonlinearity $f(y) = y^3 - y$, $f'(s) = 3s^2 - 1 \geq -1$ is bounded below, which is what makes the coercivity argument in the proof sketch available; a genuine coercivity estimate for the linearized adjoint operator additionally requires either a smallness condition on T or an exponential time-weight $e^{-c_f t}$ to absorb the -1 reaction term, which we do not carry out in detail here.

3.4 Computable A Posteriori Bound

Because $Y_\theta, U_\psi, \Lambda_\phi$ are smooth functions on all of $\overline{Q}$ (for smooth activation functions), there is no computational mesh and no notion of an inter-element jump in the normal derivative – such jumps are identically zero for a globally differentiable network. We therefore build the a posteriori indicator directly from pointwise residual evaluation on a *verification point set*, distinct from (and typically much larger/denser than) the training collocation set, rather than from a finite-element-style mesh.

3.4.1 Verification-Point Residual Indicator

Let $\widehat{Q} = \{(x_k, t_k)\}_{k=1}^K \subset Q$ and $\widehat{\Sigma} = \{(x_k^b, t_k^b)\}_{k=1}^{K_b} \subset \Sigma$ be dense verification point sets (e.g. a fixed low-discrepancy or grid sample, held out from training) with normalized quadrature weights $\{\omega_k\}$ and $\{\omega_k^b\}$, each set summing to 1. Define the global a posteriori indicator

$$\eta^2 := \sum_{k=1}^K \omega_k |\mathcal{R}_y(x_k, t_k)|^2 + \sum_{k=1}^K \omega_k |\mathcal{R}_\lambda(x_k, t_k)|^2 + \sum_{k=1}^{K_b} \omega_k^b |\mathcal{R}_u(x_k^b, t_k^b)|^2 + \eta_{\text{ic}}^2, \quad (34)$$

where η_{ic}^2 approximates $\|Y_\theta(\cdot, 0) - y_0\|_{L^2(\Omega)}^2$. The adjoint terminal mismatch is identically zero by (19). Thus η is evaluated on points independent of training and approximates the continuous residual quantity entering Theorem 1. A post-hoc partition may be used only to visualize local contributions; it introduces no jump terms.

3.4.2 A Posteriori Bound

Theorem 2 (Computable a posteriori bound). *Under Assumptions (A1)–(A3), and with η as in (34) computed on a sufficiently fine verification set,*

$$\|(e_y, e_\lambda, e_u)\|_{\mathcal{E}} \lesssim \epsilon_{\text{net}} + C_{\text{stab}}(\eta + \mathcal{E}_{\text{quad}}), \quad (35)$$

where $\mathcal{E}_{\text{quad}}$ is the (typically negligible, if K, K_b are large) quadrature error of (34) itself relative to the exact L^2 residual norms.

Sketch of proof. This follows conditionally from Theorem 1 by identifying η as a quadrature approximation of the continuous residual norm and retaining both the best-approximation and verification-quadrature contributions. $\square$

Remark 3 (Contrast with mesh-based estimators). *A finite-element a posteriori estimator is typically accompanied by a matching lower bound, $\eta(E) \lesssim \|e\|_{1,D_E}$ up to data oscillation (the “efficiency” half of the classical reliability–efficiency pair), proved with element-wise bubble functions supported on individual mesh cells. This construction relies essentially on having a mesh with compactly supported, element-local test functions. In the present meshfree PINN setting there is no such local test-function machinery available in general, and we are not aware of a proof of a matching lower bound for collocation-based PINN residual indicators of this type. This is a difference in what the two methods’ geometric structure makes available, not a deficiency of (34): η should be read as a one-sided, computable bound – useful for monitoring convergence and flagging poorly-resolved regions of Q – rather than as a certified two-sided (mesh-style) estimator. A two-sided meshfree indicator is left as an open direction.*

3.5 Practical Computation and Empirical Validation

Combining Theorem 1 with the identification of η as a quadrature estimate of ϵ_{loss} , we have informally

$$\|(e_y, e_\lambda, e_u)\|_{\mathcal{E}}^2 \lesssim \mathcal{L}_{\text{ind}} + \mathcal{E}_{\text{approx}} + \mathcal{E}_{\text{quad}}, \quad (36)$$

so a small training loss is *necessary but not sufficient* for a small true error: it must be accompanied by (i) a sufficiently expressive architecture ($\mathcal{E}_{\text{approx}}$ small) and (ii) a sufficiently dense/well-distributed collocation set ($\mathcal{E}_{\text{quad}}$ small). In particular, a low training loss evaluated *on the training points themselves* can be misleading if the network overfits those points; this is precisely why η is evaluated on an independent verification set (Section 3.4.1).

3.5.1 Algorithm

The a posteriori indicator is computed after training as follows:

1. Sample a verification point set $\hat{Q} \cup \hat{\Sigma}$, independent of the training collocation points (e.g. a fresh random draw, or a fixed low-discrepancy grid), together with quadrature weights.
2. Evaluate $Y_\theta, U_\psi, \Lambda_\phi$ and their required derivatives at these points via automatic differentiation, and compute the residuals $\mathcal{R}_y, \mathcal{R}_\lambda, \mathcal{R}_u$.
3. Form the weighted sum (34).
4. (Optional, diagnostic only) Bin the pointwise squared residuals over a post-hoc space-time partition purely for visualization of where the error is concentrated; this partition does not enter the estimator’s value.

The computational cost is comparable to one evaluation of the loss function on $K + K_b$ points, i.e. negligible relative to training.

3.5.2 Empirical validation on Examples 1 and 2

Because both numerical tests of Section 4 are manufactured-solution problems, the true error $\|e\|_{\mathcal{E}}$ against (y^*, u^*, λ^*) is directly computable, and no separate reference FEM solve is required to validate η : unlike the general setting of (30), here we can compare η against the true error on the same trained models used in Section 4. We therefore validate the indicator directly on Examples 1 and 2 as follows:

1. For each example and each collocation count $N = N_{\text{int}} = N_{\text{bc}}$, evaluate η from (34) on a verification set independent of the N training points (the same verification grid used for the residual maps of Section 4).
2. Compute the true error $\|e\|_{\mathcal{E}}$ against the manufactured solution.
3. Report the empirical ratio $\eta/\|e\|_{\mathcal{E}}$ (the *effectivity index*) as N varies, rather than assuming a fixed algebraic rate; unlike FEM, PINN convergence in N is not generally guaranteed to follow a clean power law, and any observed rate should be reported empirically.

Table 1: Effectivity index $\eta/\|e\|_{\mathcal{E}}$ on Examples 1 and 2. *Values to be filled in from the trained checkpoints of Section 4; not yet computed in this draft.*

Example	N	$\ e\ _{\mathcal{E}}$	η	$\eta/\ e\ _{\mathcal{E}}$
1	–	[TBD]	[TBD]	[TBD]
1	–	[TBD]	[TBD]	[TBD]
2	–	[TBD]	[TBD]	[TBD]
2	–	[TBD]	[TBD]	[TBD]

This validates Theorem 2 empirically on the present problems; it does not, and is not intended to, validate a lower (efficiency) bound (Remark 3). For a genuinely low-regularity or non-manufactured problem, where $\|e\|_{\mathcal{E}}$ is unavailable, the same procedure without step 2 is exactly the practical use case for which η is intended.

3.6 Summary of the Error Estimation Framework

The proposed error estimation framework provides:

- **A conditional a priori bound** (Theorem 1), valid under explicit well-posedness, regularity, and approximation-capacity assumptions (A1)–(A4).
- **A computable a posteriori bound** (34), built from pointwise residual evaluation on a verification set, with no artificial mesh-jump terms, together with an honest statement that a matching lower (efficiency) bound is not established here for the meshfree setting (Remark 3).
- **An empirical validation protocol** (Section 3.5.2), applied directly to Examples 1 and 2, reporting the effectivity index $\eta/\|e\|_{\mathcal{E}}$.

Remark 4. *The Dirichlet boundary control setting analyzed here (Assumption (A2)) is a known source of low-regularity difficulties for the classical PDE theory; results here should not be read as removing that difficulty, only as working under regularity hypotheses strong enough to bypass it. Extending the framework to genuinely low-regularity Dirichlet data, and to a two-sided meshfree a posteriori bound, are left for future work.*

4 Numerical Experiments

4.1 Common computational settings

Both tests use $\Omega = (0, 1)$, $T = 1$, $\nu = 0.1$, and $\alpha = 10^{-2}$. All networks are fully connected, use tanh activations and Xavier initialization, and are trained in double precision. A single control network $U_{\psi}(x, t)$ is evaluated at $x = 0$ and $x = 1$ to obtain the two boundary traces.

Table 2: Neural-network architectures.

Network	Input	Output	Hidden layers	Neurons/layer
Y_{θ} or N_{θ} (state)	(x, t)	scalar	4	50
U_{ψ} (control)	(x, t)	scalar	3	30
M_{ϕ} (adjoint)	(x, t)	scalar	4	50

Training uses independent uniform Monte-Carlo samples: 10 000 interior points, 2 000 points on each boundary component, and 500 initial points. Points are resampled every 500 Adam epochs. Adam is run for 20 000 epochs from learning rate 10^{-3} , with decay factor 0.9 every 1 000 epochs, followed by at most 5 000 L-BFGS iterations. The loss weights are listed in Table 3. The reported values correspond to one run with seed zero.

Table 3: Loss weights used in both experiments.

Term	Direct PINN	Indirect PINN
State residual	1	1
State boundary condition	10	hard-constrained
State initial condition	10	10
Adjoint residual	—	1
Stationarity residual	—	1
Adjoint boundary and terminal conditions	—	hard-constrained

Accuracy is reported using the root-mean-square error

$$\text{RMSE}(v) := \left(\frac{1}{N} \sum_{j=1}^N |v_j - v_j^*|^2 \right)^{1/2}. \quad (37)$$

Residual maps are evaluated on a verification grid independent of the training samples.

4.2 Example 1: cubic reaction–diffusion equation

4.2.1 Problem definition

The first test is a cubic reaction–diffusion equation with $f(y) = y^3$. We prescribe

$$y^*(x, t) = \sin(\pi x)e^{-t}, \quad u^*(0, t) = u^*(1, t) = 0, \quad (38)$$

and $y_0(x) = \sin(\pi x)$. The manufactured source is

$$g(x, t) = (-1 + \nu\pi^2) \sin(\pi x)e^{-t} + \sin^3(\pi x)e^{-3t}. \quad (39)$$

Choosing $y_d = y^*$ makes (y^*, u^*) feasible with $J(y^*, u^*) = 0$; hence it is a global optimum. The exact adjoint is $\lambda^* = 0$.

4.2.2 Direct and Indirect results

Figure 1 compares the exact state and the two PINN approximations. The direct and indirect state RMSEs are 2.34×10^{-3} and 6.80×10^{-4} , respectively. Their maximum pointwise state errors are 1.648×10^{-2} and 2.551×10^{-3} .

State comparison — Cubic reaction-diffusion test

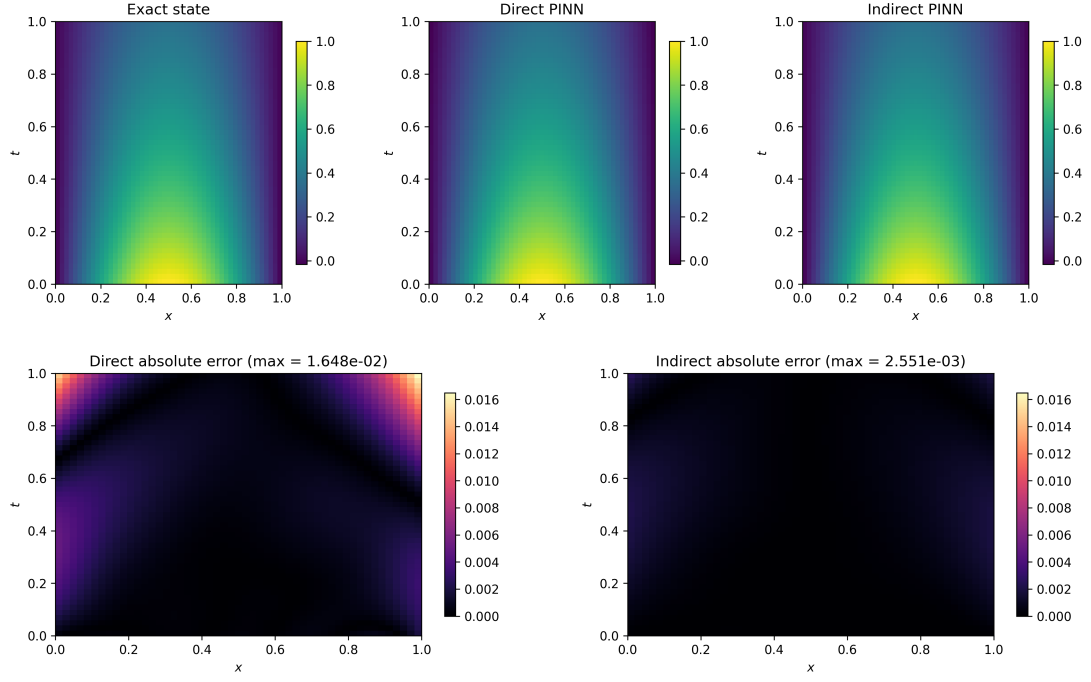


Figure 1: State approximation and absolute error for the cubic reaction-diffusion test. Both error panels use the same color scale.

The boundary-control RMSE decreases from 5.98×10^{-3} for the direct method to 1.43×10^{-3} for the indirect method; see Figures 2–3.

Boundary-control errors — Cubic reaction-diffusion test

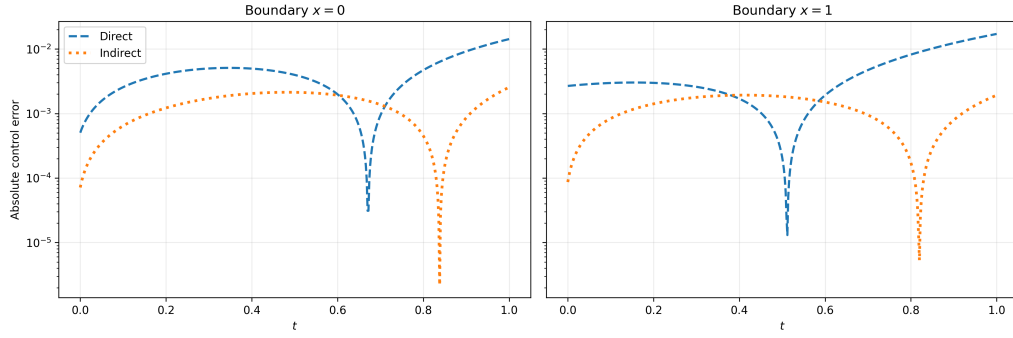


Figure 2: Absolute boundary-control errors on the left and right endpoints for Example 1.

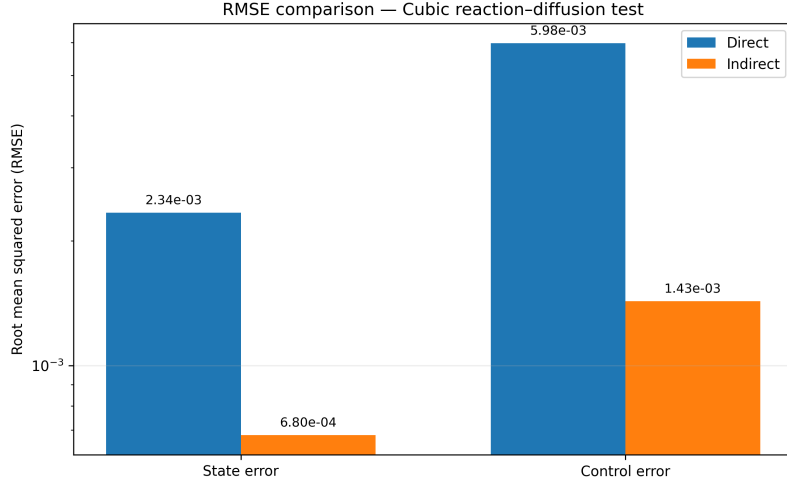


Figure 3: RMSE comparison for Example 1.

4.2.3 Discussion

The indirect formulation reduces the state RMSE by approximately 70.9% and the control RMSE by approximately 76.1%. Its largest adjoint error is 1.849×10^{-4} , consistent with the exact value $\lambda^* = 0$ (Figure 4). The verification maps in Figure 5 show generally smaller state residuals for the indirect method. The stationarity residual remains mainly of order 10^{-4} and decreases near the terminal time (Figure 6).

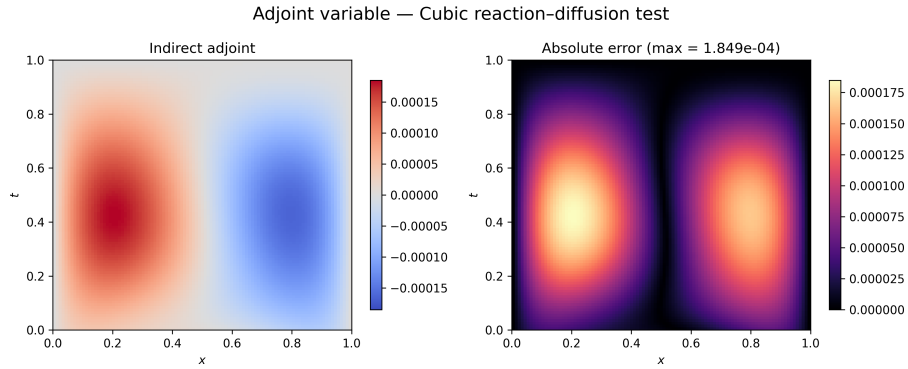


Figure 4: Indirect adjoint approximation and absolute error for Example 1.

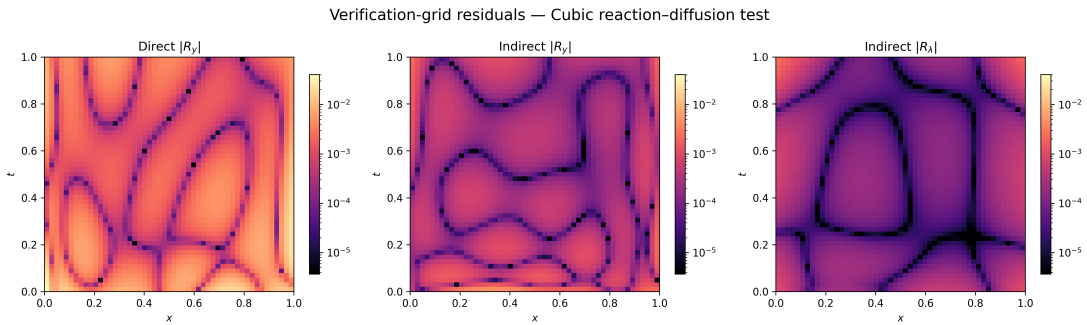


Figure 5: State and adjoint residuals evaluated on an independent verification grid for Example 1. A common logarithmic color scale is used.

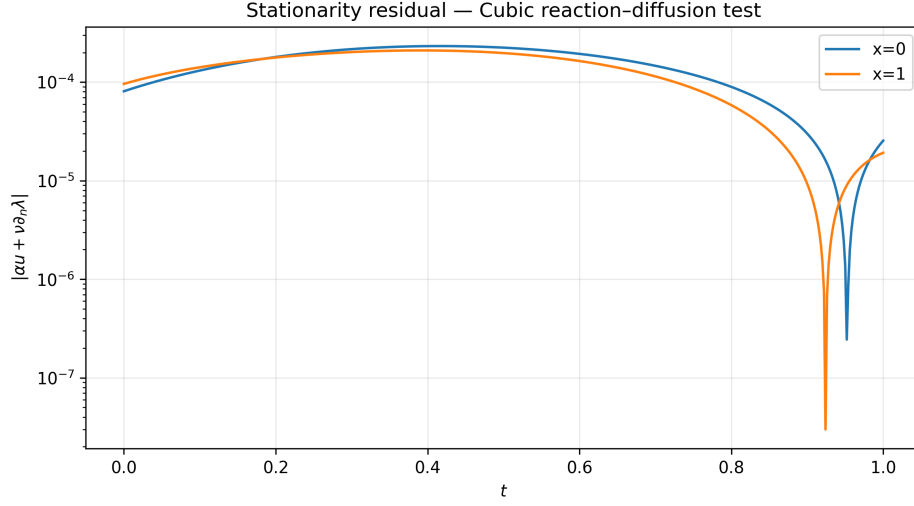


Figure 6: Stationarity residual $|\alpha u + \nu \partial_n \lambda|$ at both endpoints for Example 1.

4.3 Example 2: Linear boundary-control problem

4.3.1 Construction of the exact KKT solution

The second test uses the linear heat equation, $f(y) = 0$, and is constructed so that both the control and adjoint are nonzero. Let $A = 0.05$ and prescribe

$$\lambda^*(x, t) = A x(1 - x) \sin(\pi t), \quad (40)$$

$$u^*(0, t) = u^*(1, t) = \frac{\nu A}{\alpha} \sin(\pi t) = 0.5 \sin(\pi t), \quad (41)$$

$$y^*(x, t) = 0.5 \sin(\pi t) + \sin(\pi x) e^{-t}. \quad (42)$$

The source and desired trajectory are then defined from the state and adjoint equations:

$$g(x, t) = 0.5\pi \cos(\pi t) + (-1 + \nu\pi^2) \sin(\pi x) e^{-t}, \quad (43)$$

$$y_d(x, t) = y^*(x, t) - A\pi x(1 - x) \cos(\pi t) + 2\nu A \sin(\pi t). \quad (44)$$

The triplet (y^*, u^*, λ^*) satisfies the complete optimality system. Since the dynamics are linear and the objective is strictly convex in the control, it is the unique optimum.

4.3.2 Direct and Indirect results

Figure 7 shows close agreement with the exact state. The direct and indirect state RMSEs are 1.65×10^{-3} and 1.38×10^{-3} ; the corresponding maximum errors are 8.982×10^{-3} and 5.604×10^{-3} .

State comparison — Linear boundary-control test

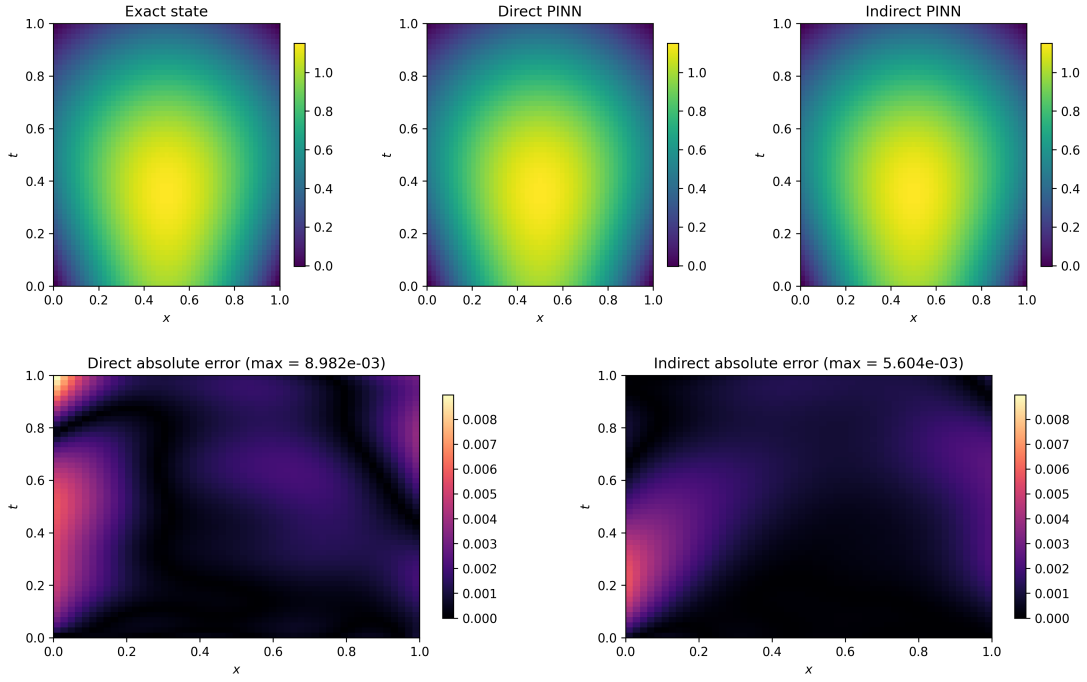


Figure 7: State approximation and absolute error for the nontrivial manufactured KKT test.

The control RMSE decreases from 4.08×10^{-3} to 2.55×10^{-3} (Figures 8–10). The indirect adjoint has maximum absolute error 6.079×10^{-4} (Figure 9).

Boundary-control errors — Linear boundary-control test

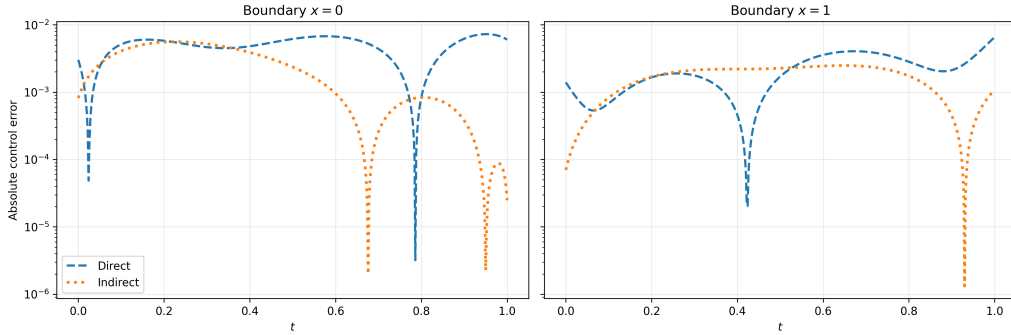


Figure 8: Absolute boundary-control errors for Example 2.

Adjoint variable — Linear boundary-control test

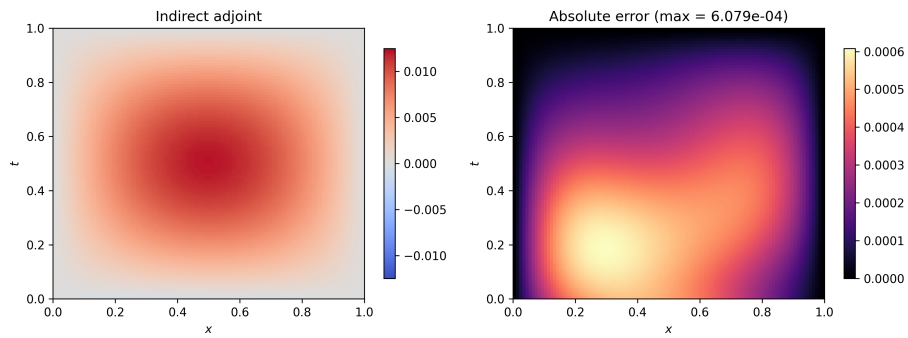


Figure 9: Indirect adjoint approximation and absolute error for Example 2.

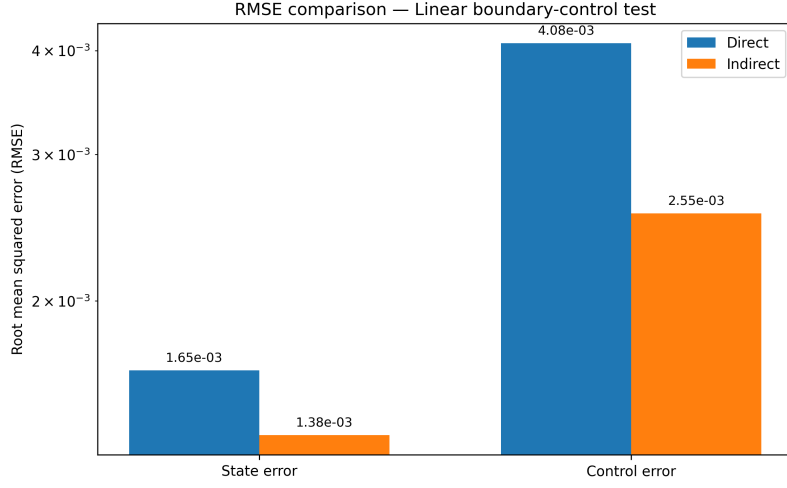


Figure 10: RMSE comparison for Example 2.

4.3.3 Discussion

The indirect method reduces the state RMSE by approximately 16.4% and the control RMSE by approximately 37.5%. Unlike Example 1, this test exercises a nonzero control, a backward adjoint equation, the terminal condition, and the stationarity relation. The control error is not pointwise smaller at every time, but its global RMSE is lower. Figures 11 and 12 show that the learned fields remain consistent with the PDE and first-order optimality conditions on independent verification points.

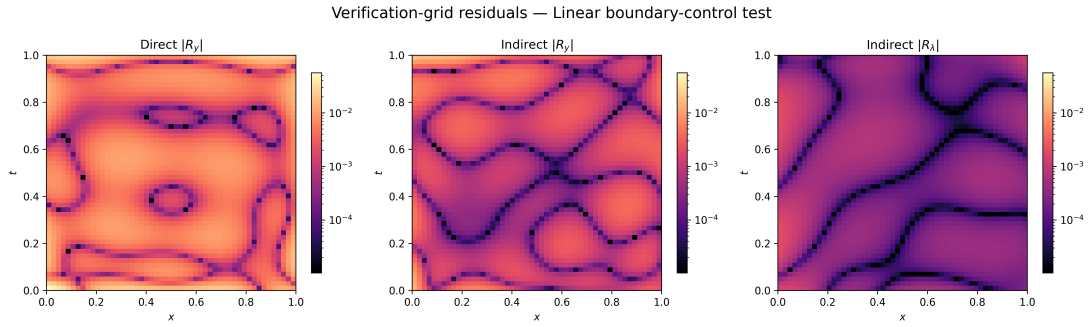


Figure 11: Verification-grid state and adjoint residuals for Example 2.

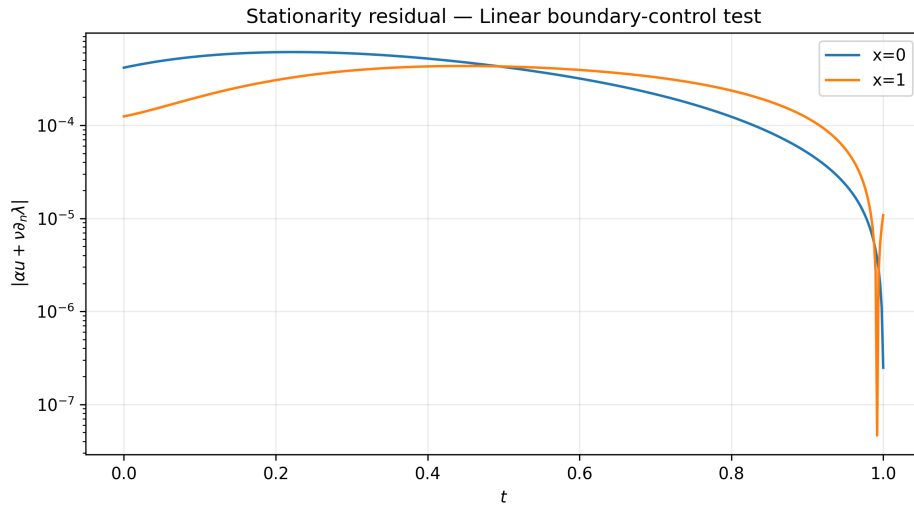


Figure 12: Stationarity residual at both endpoints for Example 2.

4.4 Overall comparison

Table 4: Summary of the numerical errors.

Example	Method	State RMSE	Control RMSE	Maximum state error
1	Direct	2.34×10^{-3}	5.98×10^{-3}	1.648×10^{-2}
1	Indirect	6.80×10^{-4}	1.43×10^{-3}	2.551×10^{-3}
2	Direct	1.65×10^{-3}	4.08×10^{-3}	8.982×10^{-3}
2	Indirect	1.38×10^{-3}	2.55×10^{-3}	5.604×10^{-3}

For the present runs, the indirect formulation gives lower state and control RMSE in both examples. The improvement is strongest in Example 1, while Example 2 confirms that the method also recovers a nontrivial boundary control and adjoint. These single-seed results indicate an advantage of the hard-constrained optimality-system formulation, but repeated runs are required for a statistical comparison.

5 Conclusions

We developed direct and indirect PINN formulations for a semilinear parabolic optimal-control problem with Dirichlet boundary control. A single network $U_\psi(x, t)$ represents the control on both boundary components. In the indirect method, a boundary lifting enforces the state-control coupling exactly, while the adjoint architecture enforces both its homogeneous boundary condition and terminal condition.

Two manufactured tests were considered. The cubic reaction–diffusion test reproduces the original zero-control setting, whereas the linear KKT test contains a nonzero control and adjoint. In the reported runs, the indirect method produced lower state and control RMSE in both examples. Independent verification-grid residuals also showed consistency with the state, adjoint, and stationarity equations.

The error estimates remain conditional on well-posedness, regularity, stability, and network approximation assumptions. In particular, no efficiency result is claimed for the meshfree residual indicator. Future work should include repeated random seeds, convergence studies with increasing collocation sets, genuinely low-regularity Dirichlet data, and nonlinear tests with nontrivial controls.

References

- [1] Z. Zhang, S. Liu, A. Alla, J. Darbon, and G. E. Karniadakis, “Pinns in pde constrained optimal control problems: Direct vs indirect methods,” *arXiv preprint arXiv:2604.04920*, 2026.